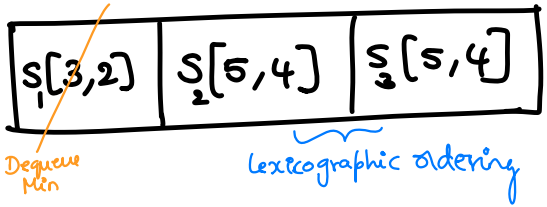
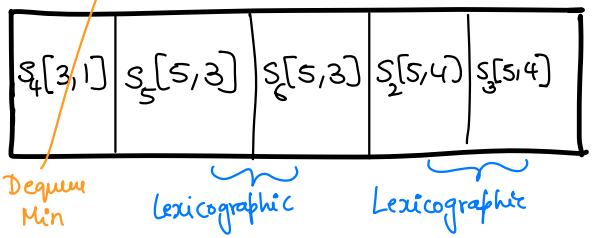
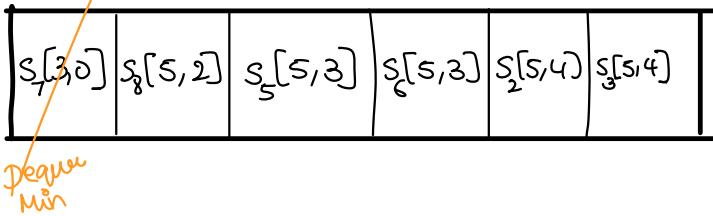
Level of Step	Dequeue-Min	Newly Enqueued States	Q after updating in lexicographic order
0	$S_0[3,3]$	$S_1[3,2]$, $S_2[5,4]$ $S_3[5,4]$	
1	$S_1[3,2]$	$S_4[3,1]$, $S_5[5,3]$ $S_6[5,3]$	
2	$S_4[3,1]$	$S_8[5,2]$, $S_7[3,0]$	
3	$S_7[3,0]$	Dequeue Min = Goal (Stop)	

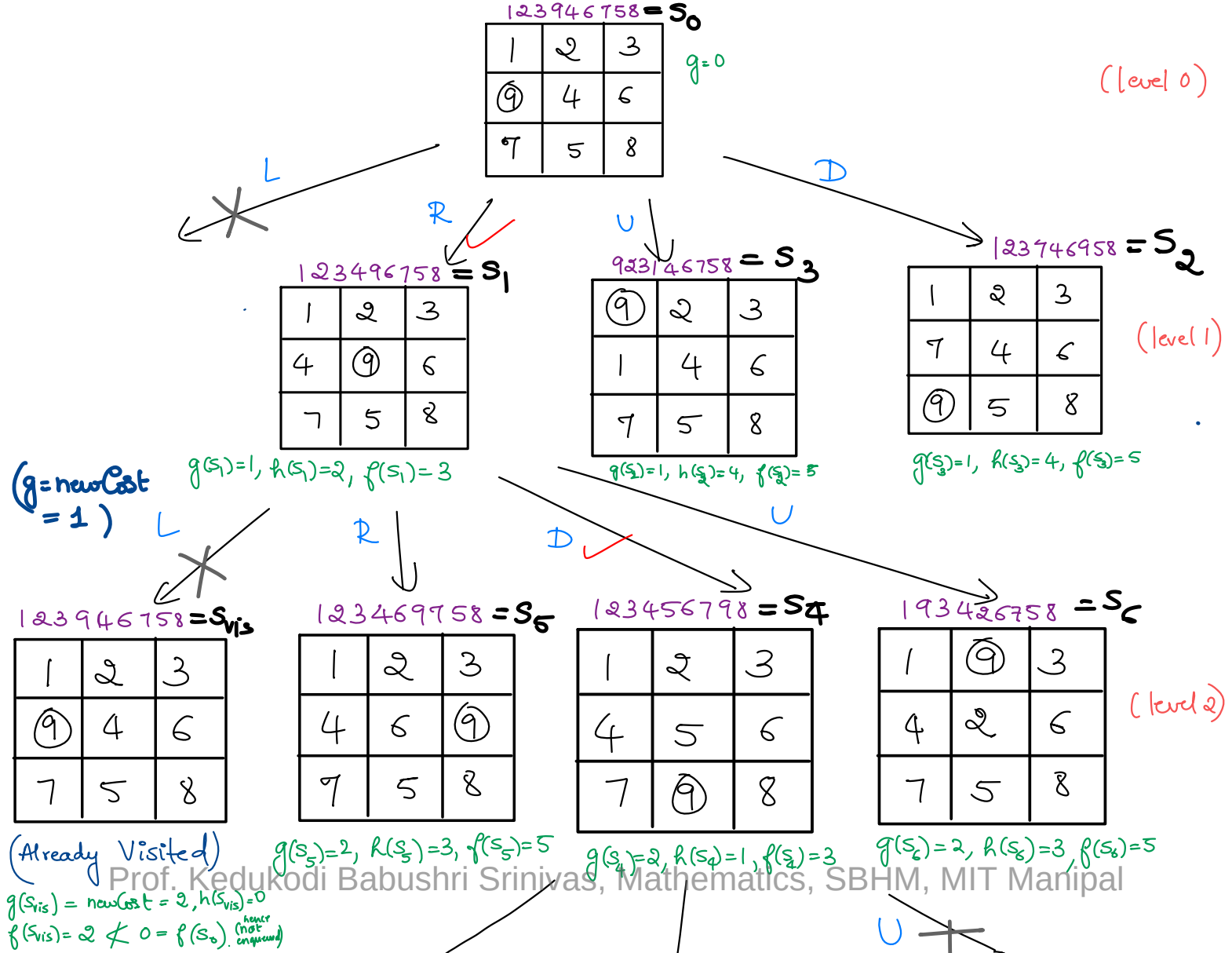
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q-table :

State	S_i	$g(S_i)$	$h(S_i)$	$f(S_i)$	Parent (S_i)	Move (S_i)	Sorted Priority Q for new nodes
123456789	S_0 (Extend)	0	3	3	Nil	Nil	$S_1[3,2]$, $S_2[5,4]$, $S_3[5,4]$
123496758	S_1 (Extend)	1	2	(3) (Min)	S_0	R	$S_4[3,1]$, $S_6[5,3]$, $S_6[5,3]$
123746958	S_2	1	4	5	S_0	D	
923146758 (Lexicographic)	S_3	1	4 (match)	5 (match)	S_0	U	
123456798	S_4 (Extend)	2	1	(3) (Min)	S_1	D	$S_7[3,0]$, $S_8[5,2]$
123469758	S_5	2	3	5	S_1	R	
193426758 (Lexicographic)	S_6	2	3	5	S_1	U	

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12345678	S_7	3	0	min 3	S_4	R	
12345678	S_8	3	2	5	S_4	L	



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L

R

$S_8 = 123456978$

1	2	3
4	5	6
⑨	7	8

$g(S_8)=3,\ h(S_8)=2,\ f(S_8)=5$

$S_7 = 123456789$

1	2	3
4	5	6
7	8	⑨

$= G$

$g(S_7)=3,\ h(S_7)=0,\ f(S_7)=3$

$S_9 = 123496758$

1	2	3
4	⑨	6
7	5	8

(level 3)

(Already Visited)

S_9

$g(S_{vis})=3,\ h(S_{vis})=2$

$f(S_{vis})=5 \not< f(S_1)=3$

Output:

$$S_0 \xrightarrow{R} S_1 \xrightarrow{D} S_4 \xrightarrow{R} S_7 = G$$

Conditions for Optimal Solution:

Let $h^*(s)$ denote the true min. remaining cost from s to the goal

① No overestimation (Admissibility)

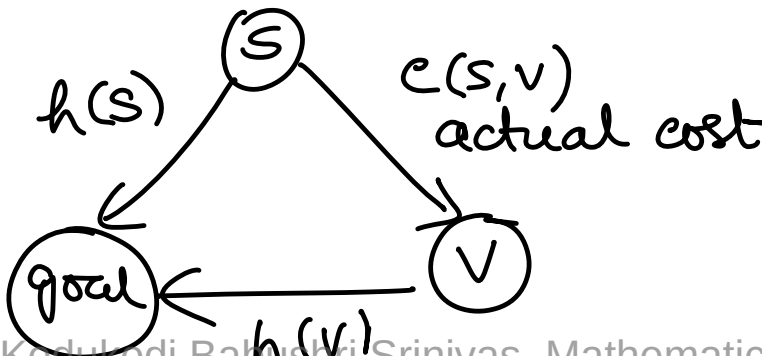
$0 \leq h(s) \leq h^*(s)$ for all states s

② Triangle Inequality (Consistency)

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For every legal transition from s to v $(s \to v)$ with actual cost $c(s,v)$ (≥ 0)

$$h(s) \leq c(s, v) + h(v).$$



Why blank 9 (d_9) should not be added in the Manhattan distance?

S_0

1	2	3
9	4	6
7	5	8

optimal solution = R, D, L

True Cost = $h^*(S_0) = 3$

$h(S_0) = d_4 + d_5 + d_8 = 1 + 1 + 1 = 3$

$h_q(S_0) = d_4 + d_5 + d_8 + d_9 = 1 + 1 + 1 + 3 = 6$

Note that $3 = h(S_0) \leq h^*(S_0) = 3$

whereas $6 = h_q(S_0) \not\leq h^*(S_0) = 3$

Considering $h_q(S_0)$, Admissibility fails

S_1

1	2	3
4	9	6
7	5	8

$$h_q(S_1) = d_5 + d_8 + d_9$$
$$= 1 + 1 + 2$$
$$= \underline{4}$$

Let $S = S_0$, $V = S_1$ $\quad C(S,V) = C(S_0,S_1) = 1$

$$h_q(S_0) \leq c(S_0,S_1) + h_q(S_1)$$
$$6 \not\leq 1 + 4 = 5$$

Triangular Inequality fails for $h_q(S_0)$

If we consider h_9,

the solution obtained using A^*

will not be optimal.

Consider $s \longrightarrow v$

$$f(v) = g(v) + h(v)$$ (From A^* algorithm)

$$= [g(s) + c(s, v)] + h(v)$$ (Extend $g(v)$ in terms of $g(s)$ using $c(s,v)$)

$$= g(s) + [c(s, v) + h(v)]$$ (Associativity of $+$)

$$\geq g(s) + h(s)$$ (because triangular inequality $h(s) \leq c(s,v) + h(v)$)

$$= f(s)$$

Hence $f(v) \geq f(s)$,

that is, f is nondecreasing along every path. $\equiv$